On the Effects of Pre- and Post-Processing in Video Cartoonization with Bilateral Filters

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Fig. 1. Various cartoonization outputs of a sample video frame from the movie *Mr. Bean's Holiday* (top left). The following schemes that utilize pre- and post-processing are shown: CL+ADJ (top right), CL+ADJ+LHPF (bottom left), CL+ADJ+US (bottom right).

***Abstract*—In recent years, advances in image-based artistic rendering have grown steadily with the additional leverage of image and video processing techniques. Video cartoonization or stylization is the process of artificially incorporating cartoon-like effects to photorealistic input videos. This paper investigates the effects of integrating relevant pre- and post-processing tasks to significantly improve the quality of cartoonized videos processed with bilateral filters (BF). Our video cartoonization framework work extends the original Winnemöller's real-time video abstraction framework, which applies the edge-preserving BF with additional use of edge maps, luminance quantization and frame temporal coherency. In our work, we propose a contrast enhancement option by intensity stretching and Laplacian filtering to finetune the contrast levels of the pre-BF frames. For the post-BF recombined frames, an unsharp masking procedure is proposed to accentuate feature details in the final output video. Results from extensive experiments conducted by qualitative user evaluation underline the essentiality of pre- and post-processing tasks for improved video cartoonization.**

I. Introduction

The field of non-photorealistic rendering (NPR) has rapidly developed into a popular area of research which covers many aspects of visual information processing and multimedia applications. Interestingly, the convergence of computer graphics and image/video processing has enabled the artistic stylization of images/video to be accomplished using various image/video processing techniques and computer vision algorithms, unlocking many new possibilities.

Image and video stylization commonly refers to stylized rendering of images or video to incorporate non-realistic cartoon-like effects to the output. Other equivalent phrases include *video tooning* [1] and *video abstraction* [2, 3] while the term *cartoonization* is used in this paper to refer to the somewhat similar task, but in a more output-centric manner. Cartoon-like animations can be artificially created from a photorealistic input video, such as movies, documentaries or self-captured videos. A common approach to create non-photorealistic cartoon-like depictions is by transforming the images or video using a set of filtering techniques or rendering styles that can be either interactive or automatic.

In a new taxonomical survey by Kyprianidis et al. [4] on the state-of-the-art artistic stylization techniques for images and video, two complementary trends in NPR literature were mentioned. Firstly, NPR can be fused with higher level computer vision, trending towards complex analysis of scenes to incorporate artistic abstraction. Secondly, there has been a growing interest in the use of local processing approaches, especially in the area of edge-aware filtering for real-time

stylization of images and video. The scope of this paper falls within this second motivation, whereby the use of classic bilateral filtering for video cartoonization is thoroughly investigated and evaluated in the light of additional local processing procedures. Instead of focusing on simulating a particular artistic technique or style, the process of image abstraction or cartoonization revolves around simplification of realistic scene information by removing information that are irrelevant to cartoons, while accentuating information that project more cartoon-like features.

In this paper, we investigate the significance of proposing relevant pre- and post-processing tasks that are essential towards the improvement of video cartoonization by bilateral filters (BF). Our work extends the original video abstraction framework proposed by Winnemöller et al. [3], which performs abstraction by artificially altering the low and high contrast regions in addition to edge-preserving bilateral filters to obtain cartoon-like effects with good temporal coherency. Contrast enhancement by intensity stretching and Laplacian filtering further tunes the contrast of the pre-BF frames, while an unsharp masking procedure is proposed for post-BF recombined frames to heighten the feature details in the stylized output video. Experiments were conducted by user evaluation and the results were thoroughly presented using a set of qualitative measures.

The rest of this paper is organized as follows. Section II reviews some related works in brief. Section III describes the video cartoonization framework used and the modules involved. A detailed elaboration of our experimental setup and results based on a set of qualitative measures are shown in Section IV, with some further analysis. Finally, conclusions and future directions are provided in Section V.

II. Related Work

In an early work, video stylization was first addressed by Litwinowicz [5], where "painterly" videos are produced by pushing brush strokes from frame to frame in the direction of optical flow motion vectors. This approach was extended by Hays and Essa [6] who similarly move strokes within independent motion layers, while encoding spatio-temporal variations for rendering effects of selected painterly styles. A later work by Bousseau et al. [7] harnessed optical flow to advect strokes between adjacent frames. The exploitation of video properties have already gained much traction, with Hertzmann and Perlin [8] using the differences between consecutive frames of video to paint over areas of the new frame that differ significantly from the previous frame. While these methods produced impressive painterly videos, the errors in the estimated per-pixel motion field can quickly accumulate and propagate to subsequent frames, resulting in increasing temporal incoherence. Strokes of the stylized output may no longer match object motion, causing a "flickering" phenomenon.

Some works are particularly motivated to utilize image processing and computer vision techniques. DeCarlo and Santella [9] employed the Canny edge detector and mean-shift filter to obtain a cartoon-style image abstraction. The edge detector is applied to produce line drawing, while the mean-shift filter performs region smoothing and segmentation. An eye-tracking user interface was also implemented to allow user-guided specification of regional importance that enables adaptive control of the level of abstraction. This approach was improved by Wang et al. [1] who proposed a video tooning system for transforming an input video into a highly abstracted, spatio-temporally coherent cartoon animation. An anisotropic mean-shift filter was applied for region abstraction while users are also allowed to choose the final cartoonization style of the video. Collomosse et al. [10] similarly used the mean-shift filter to solve an offline video abstraction problem by achieving good spatio-temporal coherence.

There are multiple limitations in some of these earlier methods. The approach by Winnemoller et al. [3] is a big step of improvement over the earlier methods. A fully automatic pipeline for the stylization of cartoon renderings was introduced, such that images and videos can be stylized by changing the contrast of visually important features such as edges, luminance and color opponency. Their landmark pipeline employs the bilateral filter (BF) [11] and Difference-of-Gaussians (DoG) filter, bringing forth several influential ideas that other researchers later built upon. Although computation of the bilateral filter can be comparatively expensive, but an alternative separable implementation by Pham and van Vliet [12] that uses a numerical scheme can be applied iteratively to good effect. In addition, smooth quantization of the luminance channel in *CIELab* colorspace is performed to achieve an artistically attractive cartoon-like effect with good temporal coherence. The choice of the bilateral filter is justified for its effectiveness in smoothing regions and preserving edges in the presence of noise. In a new extension, Winnemöller [13] showed the versatility of the DoG filter and how a large variety of NPR effects can be created by properly adjusting its parameters.

Kang et al. [14] presents a NPR feature technique that automatically performs line drawing from a photograph by extracting a set of coherent, smooth, and stylistic lines that effectively capture and convey important shapes in the image. In addition, an edge tangent flow and a flow-based DoG filter are applied to improve the lines drawn on the rendered images The avenues for video cartoonization applications are immense. Fischer et al. [15] proposed to use an automatic stylization method for augmented reality, whereby stylization is applied to both virtual and real inputs to fuse virtual objects into a live video stream. The Canny edge detector and bilateral filter were used for line extraction and region smoothing respectively. In another recent work, Wang et al. [16] presented a Digital Ambient Display (DAD) that harnesses artistic stylization to create an abstraction of users' experiences through their home video collections. The DAD automatically selects, stylizes and transitions between clips enabling users to passively consume their video collections and rediscover past memories. Temporally coherent regions were stylized via either shading or stroke-based rendering, to produce coherent cartoon and painterly video styles.

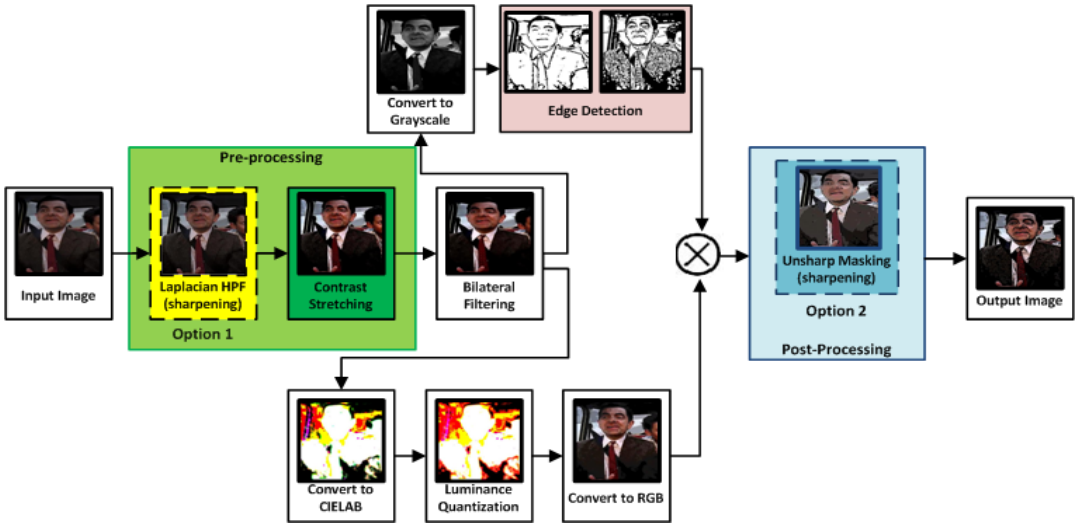

Fig. 2. Framework diagram for the proposed image cartoonization pipeline with pre- and post-processing modules included. Note that the blocks with dashed borders indicate optional modules for the purpose of our experimental evaluation.

III. Video Cartoonization Framework

Our work is built upon the video abstraction framework proposed by Winnemöller [3], which has since been established as the generalized pipeline for cartoonization by way of local filtering [4].

We further extend Winnemöller's original framework to incorporate pre- and post-processing modules, as shown within the image cartoonization pipeline in Figure 2. In this framework, we propose the use of Laplacian high pass filter (LHPF) and unsharp masking procedure to complement the main cartoonization step. The main purpose of these two tasks is to accentuate the sharpening of features and image details that will eventually strengthen the effects of bilateral filtering. These two steps are regarded as "options" in the framework as they can be plugged in (or removed) to adjust the effects on the output video, which is essentially useful in our experimentation in Section IV. In addition, contrast stretching is selected as a fixed step in the pre-processing module of our framework to enhance the image contrast while maintaining overall balance in intensity range.

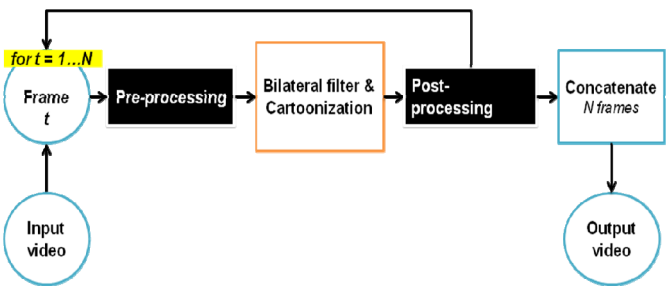

Fig. 3. High-level flowchart of the proposed video cartoonization framework

In order to realize a complete video cartoonization task, the video should be first converted to individual image frames, and the cartoonization process is applied to each frame. At the end of the process, all cartoonized frames will be combined together to form the final cartoonized output video. The high-level diagram of the video cartoonization framework is shown in Figure 3. We briefly discuss the modules involved in the following sub-sections.

A. Bilateral Filtering

Bilateral filtering smoothens images while preserving edges with a nonlinear combination of nearby image values, a concept closely resembles anisotropic diffusion. The key idea of the bilateral filter is to apply Gaussian-based weights to neighbouring pixels in attempt to obtain spatial smoothing. To yield good stylized and abstracted result, an iterative bilateral filter was suggested by Tomasi and Manduchi [11]. Concretely, the bilateral filter, denoted by BF, is defined by:

$$BF[I]_p = \frac{1}{W_p} \sum_{q \in S} G_{\sigma_s}(\|p-q\|) G_{\sigma_r}(|I_p - I_q|) \quad (1)$$

where I_p and I_q denote the image value at pixel position p and q respectively while the L2-norm term denotes the Euclidean distance between the two pixels. The normalization factor W_p ensures pixel weights sum to 1.0,

$$W_p = \sum_{q \in S} G_{\sigma_s}(\|p-q\|) G_{\sigma_r}(|I_p - I_q|) \quad (2)$$

G_{σ_s} defines the *spatial* Gaussian weights that compute the influence of distant pixels while G_{σ_r} defines the *range* Gaussian weighs that compute the influence of pixel q when their intensity values differ from I_p. The intensity of a bilateral filtered pixel at position p, $BF[I]_p$ is summed from over all

possible spatial locations define in neighbourhood set *S*. In our framework, the bilateral filter is applied directly to the original RGB colorspace, an improvement in processing speed since an explicit conversion to the *CIELab* colorspace is not required. The bilateral filtered image will be used in both the luminance quantization and edge detection steps thereafter.

B. CIELab Colorspace Conversion

Tomasi and Manduchi [11] recommended computing the bilateral filter on a perceptually uniform feature space such as CIELab to allow image contrast to be adjusted based on noticeable differences. By natural design, CIELab approximates the human vision by describing all the colours visible to the human eye. Its '*L*' luminance component closely matches human perception of lightness, while '*a*' and '*b*' are for color-opponent dimensions. In our framework, the bilateral filtered output image is converted to CIELab colorspace before luminance quantization is performed. This slight alteration from Winnemöller's allows the luminance properties to be derived from the region smoothened image.

C. Luminance Quantization

In the first of the two abstracted features, color quantization or luminance quantization is used to stylize images to create cartoon-like regions with good temporal coherency. We apply the luminance quantization technique used in [3], which is defined by the following equation:

$$Q(\hat{x}, q, \varphi_q) = q_{nearest} + \frac{\Delta q}{2} \tanh\left(\varphi_q .(f(\hat{x}) - q_{nearest})\right) \quad (3)$$

where Q is the pseudo-quantized image, Δq is the bin width, $q_{nearest}$ is the bin boundary closest to $f(x)$, and φ_q is a parameter controlling the sharpness of the transition from one bin to another and is useful to minimize jarring transitions.

D. Edge Detection

The second abstracted feature is the edge map. Generally, edges are defined by high local contrast, so by adding visually distinct edges to regions of high contrast, we can further increase the visual distinctiveness of these locations. Given the importance of edges in popular art styles such as line drawings or sketches, many stylization systems employ an edge detector as part of their processing pipeline. In our work, we consider two edge detection methods:

1) Difference-of-Gaussian (DoG) filter method: The DoG operator has been shown to yield aesthetically pleasing edge lines without requiring much post-processing, particularly in the case of producing line drawings and cartoons. DoG edges are computationally cheap and are not prone to disconnectedness but they may drift from the true edges for large values of σ (of the Gaussian distribution). Besides, the thickness of the edges also scales naturally with the value of σ, providing us with the flexibility in stylizing edges.

2) Coherent Line Drawing method: Line drawing is an extreme case of classic edge detection, where all the pixel colors except at edges are "flattened down" to the same background level (white). Kang et al. [14] introduced this non-photorealistic rendering technique that automatically generates a high-quality line drawing from a photograph, where meaningful structures in the scene are captured and displayed with a set of coherent, smooth and stylistic lines with little or no clutter. This method constructs a smooth direction field that preserves the flow of the salient image features while suppressing unwanted noises. This remains the primary advantage over the DoG method where obtaining true edges may be less accurate. More details of this method can be found in the authors' original paper [14].

E. Pre-processing Tasks

We present the following pre-processing tasks that are applied to finetune the contrast levels of the pre-BF video frames:

1) Laplacian High-Pass Filtering (LHPF): In our design, the input image can be passed through this optional LHPF step to sharpen the image. This acts to highlight the strongest edges and to compensate the over-smoothing effect caused by bilateral filter in the proceeding step. We implement it using the following 3×3 mask:

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (4)$$

2) Contrast Stretching (ADJ): This fixed pre-processing step is applied to improve the contrast in an image by linearly stretching the range of intensity values it contains to span a desired range of values

$$p' = (p - I_{\min})\left(\frac{u - l}{I_{\max} - I_{\min}}\right) + l \quad (5)$$

Pixel p is transformed to p' by performing contrast stretching where I_{min} and I_{max} are the minimum and maximum intensity values of the image, while u and l are the upper and lower bounds of the new intensity range. This step can artificially increase the dynamic range of the image, imposing a highlighting effect.

F. Post-processing Task: Unsharp Masking (US)

In the post-processing module of our framework, we propose to apply unsharp masking on the post-BF recombined frames to accentuate feature edges of object without further increasing noise. Unsharp masking is a process of sharpening images by subtracting a blurred/smoothened version of an image from the image itself. It is expressed simply in two

steps— the first computes the edge image E_s, while the second step performs the masking, or combining of the edge image back into the original image:

$$E_s(x,y) = I(x,y) - \hat{I}(x,y) \tag{6}$$

$$I_{sharp}(x,y) = I(x,y) + E_s(x,y) \tag{7}$$

where I_{sharp} denotes the sharpened image obtained by unsharp masking and $\hat{I}(x,y)$ is a blurred version of $I(x,y)$.

IV. Experimental Evaluation

We conduct our experiments using 5 selected movie clips extracted from three movies: "Charlie St. Cloud", "Mr. Bean's Holiday "and "Gulliver's Travels". In the evaluation process, 15 participants (8 males and 7 females with their ages ranging from 18 to 67 years) were given 5 sets of videos (one for each move clip), each set containing 6 different cartoonized versions of the same clip. These versions were randomly ordered to eliminate bias. Seven of the participants have reasonable expertise in the art and graphics background while the rest are novices. Qualitative assessment of the cartoonized videos was performed by obtaining user feedback ratings on the degree of attractiveness of the videos. The six cartoonization techniques applied to each video are listed in Table I with their respective abbreviations. Some sample cartoonized frame results are shown in Figure 1 while Figure 5 shows selected cropped scenes from the tested movie clips.

TABLE I
List of Evaluated Cartoonization Methods

Method #	Abbreviation	Description
1	**WINN**	Winnemöller's [3] framework
2	**KANG**	Kang's [14] framework
3	**CL+ADJ**	Proposed framework: Coherent line, contrast stretching.
4	**CL+ADJ+US**	Proposed framework: Coherent line, contrast stretching and unsharp masking.
5	**CL+ADJ+LHPF**	Proposed framework: Coherent line, contrast stretching and Laplacian high pass filtering.
6	**DOG+ADJ+US**	Proposed framework: DOG edges, contrast stretching and unsharp masking.

A. Qualitative Evaluation Criterion

In order to substantiate the outcome of our subjective assessment, we propose the use of three different evaluation criterions to verify the consistency of our results. For all criterions, the total score or votes received by each cartoonization method is determined by summing up the score S for V evaluated movie clips, for U number of participants.

$$\text{Total score/vote, } \hat{S} = \sum_{u=1}^{U}\sum_{v=1}^{V} S \tag{8}$$

Figure 4 shows the results (total score/vote) of the evaluated cartoonization methods, using the following 3 evaluation criterions:

1) Criterion 1 – Normalized Scoring: The user gives a numerical score from 0 to 100 for each evaluated video clip. We perform *min-max normalization* of the scores across all evaluated methods as a measure of relative attractiveness. The normalized score, S_{norm} can be expressed as:

$$S_{norm} = \frac{S - S_{\min}}{S_{\max} - S_{\min}} \tag{9}$$

where $S_{\min}$ and $S_{\max}$ are the minimum and maximum scores among the evaluated methods.

2) Criterion 2 – Winner-takes-all Voting: The user selects the most attractive cartoonized version of the video clip, which will be given a single vote (1 vote) while the rest receive no votes.

3) Criterion 3 – Point-based Voting: The user gives a numerical score from 0 to 100 (similar to Method 1), and the corresponding points are assigned to points from 50 to 100 according to stepwise intervals. Precisely, scores 0 to 40 receive 0 point while scores 50 to 100 receive 1 to 6 points as shown in Table II.

TABLE II
Voting Point Conversion for Criterion 3

Scores	0-40	50	60	70	80	90	100
Points	0	1	2	3	4	5	6

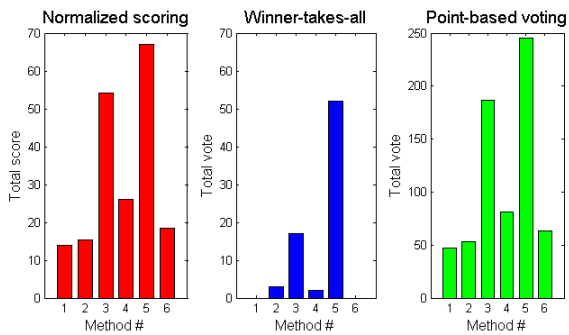

Fig. 4. Qualitative evaluation results (total score/vote) of the cartoonization methods based on the three evaluation criterions.

B. Analysis of Results

By observing the bar diagrams in Figure 4 as a whole, it is clear that a majority of users are unanimous in rating method #5 (CL+ADJ+LHPF) as the best cartoonization scheme compared to the five other methods. This firmly demonstrates the importance of pre-processing in a "local filtering" cartoonization pipeline, an appealing option which does not feature in previous works. From all three criterions, the CL+ADJ+LHPF is seen as an improvement over the CL+ADJ method as the Laplacian filter provides an initial strengthening

of edge features before the edge map is extracted. The winner-takes-all criterion shows a larger extent of this improvement when users are only allowed to pick the best video.

By comparing methods #4 (CL+ADJ+US) and #6 (DOG+ADJ+US), the choice of edge detection method is also essential in producing good cartoonized effects. The coherent line method appears to be more appealing than the DOG method originally proposed in Winnemöller's framework. Interestingly, the additional post-processing step did not seem to contribute positively to the versions that already have the pre-processing step included. This is possibly due to wrong selection of method used (unsharp masking) that led to a deterioration in attractiveness. Generally, one could observe the difference pre- and post-processing makes to the overall attractiveness of the cartoonized videos. The two original frameworks (methods #1 and #2) performed rather poorly compared to the enhanced versions.

V. Conclusion

Concretely, this paper presents the use of relevant pre- and post-processing tasks that essentially improve the quality of video cartoonization using bilateral filters (BF). By extending Winnemöller's real-time video abstraction framework, the effect on the cartoonized output was examined thoroughly. This was accomplished by conducting an extensive qualitative user evaluation that utilizes three different criterions for measuring the attractiveness of the cartoonized videos. In our modified framework, enhancement by contrast stretching and Laplacian high-pass filtering was proposed to finetune the contrast levels of the pre-BF frames. In the post-BF step, the recombined frames were put through an unsharp masking procedure to accentuate feature details in the final output video. Experimental results demonstrated some promising results, highlighting the usefulness of pre- and post-processing steps in improving the output of video cartoonization.

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Fig. 5. Sample cropped scenes from the evaluated movie clips cartoonized with CL+ADJ (right side images) and CL+ADJ+LHPF (left side images).